A) **ANCHOR SCENES (3–6)**

- **SCENE #1: First Tropane Synthesis and Cocaine Structure**

Summary: In 1901 Willstätter achieved the first total synthesis of tropinone (the bicyclic core of tropane alkaloids) from cycloheptanone, and from it elucidated the structure of cocaine [1]. This multistep synthesis (overall yield ~0.75%) demonstrated that cocaine's core structure is the tropane skeleton, resolving a long-standing uncertainty.

Date/Place: 1901, University of Munich (under mentor Adolf von Baeyer).

People: Richard Willstätter and his Munich laboratory team; Adolf von Baeyer (advisor) was nearby, though this work was primarily Willstätter's.

Concrete details: He started from readily available cycloheptanone, elaborating a cumbersome sequence of many steps [1]. The final product matched natural cocaine (structure confirmed by comparison), making it "the first synthesis and elucidation of the structure of cocaine" [2]. **Why it matters:** This established for the first time the carbon framework of a major alkaloid (cocaine) and demonstrated the feasibility of complex alkaloid total syntheses. It foreshadowed later biomimetic syntheses (e.g. Robinson's one-pot tropinone, 1917) and underscored the porosity between natural products and synthetic chemistry.

Evidence quality: Strong primary/peer-reviewed. Direct chemical publications and modern retrospectives (e.g. Mukesh Doble 2007) document this milestone [1].

Myth risk: Low. The accomplishment is well documented (also noted on Wikipedia) and not a romanticized anecdote.

Sources: Mukesh Doble (2007) [1]; Tropinone chemistry overview [2].

- **SCENE #2: Isolation of Anthocyanin from Blue Cornflower (Cyanin)**

Summary: In 1913 Willstätter (with student A.E. Everest) isolated "cyanin", the blue pigment of cornflower, and showed it was the same anthocyanin found in red roses [3]. He determined its structure as the glycoside of a flavylium salt. Crucially, he observed that the color shift (red vs. blue) was due to pH: in acid the pigment is red, in base it turns blue.

Date/Place: 1913, Kaiser-Wilhelm-Institut für Chemie (Berlin-Dahlem).

People: Richard Willstätter, Arthur E. Everest (research assistant), and lab colleagues. **Concrete details:** Willstätter extracted the pigment with neutral solvents and crystallized the red oxonium salt of cyanin [3]. He noted its remarkable color change with pH ("red in acidic, blue in alkaline solution" [3]). This was a major chemical breakthrough, revealing that flower color could hinge on simple acid–base chemistry.

Why it matters: This was the first definitive isolation of a purified anthocyanin. It explained why cornflowers and roses, despite having the same pigment, appear different colors. It laid groundwork for understanding plant flavonoid pigments and the chemistry of pH-dependent chromophore shifts in biology.

Evidence quality: Strong. The discovery is reported in contemporary chemical literature and later reviewed by chemists (Takeda 2006) [3]. The experiment was carefully documented. **Myth risk:** Low. Details (year, plant, collaborator) are precise and confirmed by primary sources. Everest later disputed the pH-only explanation, but the basic finding of cyanin's identity is solid. [3][4]

Sources: Takeda 2006 (Japan Academy) [3]; Nobel biography [4].

- **SCENE #3: Decoding Chlorophyll – Magnesium in the Green Pigment**

Summary: Between 1906 and 1914 Willstätter dissected chlorophyll chemistry to its core. He showed

that all green leaves contain essentially the same chlorophyll, a mixture of two compounds (now [5] called chlorophyll *a* and *b*) . He hydrolyzed chlorophyll to crystalline derivatives (chlorophyllin and phaeophytin) and proved that *chlorophyllins* are magnesium complexes analogous to hemoglobin's [6] [5] iron porphyrin . This work won him the 1915 Nobel Prize in Chemistry.

Date/Place: ~1909–1914, KWI Berlin-Dahlem.

People: Willstätter, co-workers (e.g. Arthur Stoll), and colleagues at KWI.

Concrete details: Using acid and base to split leaf pigment, he isolated a stable magnesium-free [6] derivative (phaeophytin) and a magnesium-containing fraction (chlorophyllin) . He measured ~3% [6] Mg in chlorophyllins, bound in a porphyrin ring ("as haemoglobin contains iron"). He also found a [7] long phytol ester chain (by ozonolysis experiments). These findings linked chlorophyll's structure to its function in photosynthesis.

Why it matters: Willstätter's elucidation of chlorophyll's chemistry laid the foundation for modern biochemistry of photosynthesis. Identifying Mg in chlorophyll was a milestone connecting plant photopigments to biophysical function. These discoveries had vast scientific impact (e.g. enabling artificial photosynthesis research) and defined his career (Nobel Prize).

Evidence quality: Very strong. His Nobel lecture and prize citation explicitly credit this work [8] [6] . Primary publications and historical reviews (Nobel archives) document the methods and conclusions. **Myth risk:** Low. The general story (Nobel Prize for chlorophyll structure) is well known and well sourced. There are no sensational anecdotes; details come from Willstätter's own publications and Nobel archives.

Sources: Nobel Prize facts and biography [5] ; Nobel Lecture (1920) [8] [6] ; MPIC history of KWI Berlin .

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SCENE #4: Resignation from University of Munich (1924)

Summary: In 1924, as anti-Semitic agitation grew on campus, Willstätter abruptly resigned his professorship in Munich. Right-wing posters demanded "No German youth may sit at the feet of a Jewish teacher," and Willstätter responded by writing a letter to the dean: *"I humbly ask your* [9] *Spectability to… request that I be dismissed… My decision is final."* This principled act ended his academic career in Germany.

Date/Place: 1924, University of Munich (Munich city).

People: Willstätter; Munich University Academic Senate; students (who later demonstrated in support); dean of faculty.

Concrete details: The resignation letter (cited above) was addressed to the university dean. It immediately caused an uproar: students staged sympathetic demonstrations, and colleague Emil [10] Kraepelin famously quipped, "He should have stayed and thrown everyone else out" . Willstätter, however, accepted his self-imposed exile in Munich, living quietly and only consulting on patents from then on.

Why it matters: This was a decisive human and career turning point. It illustrates Willstätter's moral stance against anti-Semitism and marks the premature end of his official scientific work in Germany. It also foreshadows the even darker fate facing Jewish scientists.

Evidence quality: Strong. The resignation is documented by official correspondence and [11] contemporary news, and is cited in multiple biographies . We have his own words from the letter (primary evidence).

Myth risk: Low. While dramatic, this incident is well recorded (letter text and press coverage) and not 2

an unfounded legend.

Sources: DPMA historical profile [9] ; MPIC chronology [12] ; Nobel biography [13] .

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SCENE #5: Escape from Nazi Arrest (1938)

Summary: On November 10, 1938 (the night after Kristallnacht), Gestapo agents came to arrest Willstätter at his home in Munich. By fate, he was in his garden at the time, so he slipped away [14] unnoticed . Realizing he was soon at risk, he fled Germany. After an abortive attempt via Lake [15] Constance, Willstätter finally escaped with the help of loyal former students . He left behind almost all possessions (library, Nobel medal, art collection) and found refuge in Switzerland. **Date/Place:** Nov 1938 (Munich) and early 1939 (flight to Switzerland).

People: Willstätter; Nazi Gestapo officers; his wife Sophie (Jewish) and family; students (notably Arthur Stoll) who arranged his exit.

Concrete details: According to accounts, Gestapo arrived the morning after the pogroms; [15] Willstätter was "in his beloved garden" so they did not find him . In desperation he hurriedly packed and attempted to cross into Switzerland via Bodensee (Lake Constance). That failed, but his former student A. Stoll then secured transit papers. He finally settled in Muralto (near Locarno) in March 1939, grieving the loss of his life's work.

Why it matters: This escape ensured Willstätter's survival; it was the capstone of his life story amid Nazi persecution. It also underscores the human cost of exile (abandoning scientific legacy). For science, it meant Germany lost one of its greatest chemists.

Evidence quality: Strong. Trauner's biography and DPMA account both recount the event with [15][16] eyewitness detail . Willstätter later described these days in his autobiography. **Myth risk:** Low. The dramatic escape is corroborated by multiple historical accounts. The timing and [15] circumstances (garden, Gestapo) appear in scholarly biographies .

Sources: Trauner (2015) [17]; DPMA profile [16].

B) **DIRECT QUOTES FROM WILLSTÄTTER (3–7)**

"The intention behind my work was to establish the constitutional characteristics of the most widely distributed plant pigments, of chlorophyll in particular, and to gain some criteria with regard to its chemical function." [18]

Context: Nobel Lecture (Stockholm, 1920). Willstätter describes his goal in studying plant [18] pigments and chlorophyll. *Source:* Nobel Prize Lecture in Chemistry 1915 (published 1920) .

"Analysis showed that the chlorophyllins are magnesium compounds. They contain the metal in an electrically non-dissociable state, as haemoglobin contains iron…" [6]

Context: Same Nobel Lecture. He reports his key finding that chlorophyllin derivatives contain magnesium tightly bound in a porphyrin ring, analogous to iron in hemoglobin. [6] *Source:* Nobel Prize Lecture (1920) .

"I humbly ask your Spectability to pass on to the State Ministry for Education and Culture… the request that I be dismissed from the Bavarian Civil Service. … My decision is final," he [9] wrote to the dean .

Context: Letter of 1924 (resignation from Munich University). Willstätter's own words in protesting campus anti-Semitism. *Source:* DPMA historical account quoting Willstätter's [9] resignation letter .

"I left my beloved German homeland, which had been everything to me, and found 'toleranza per riposo' as an emigrant to Muralto-Locarno, Villa Eremitaggio." [19]

Context: Final passage of his autobiography (translated into German). Reflecting on exile in Switzerland at the end of his life (c.1939–42). *Source:* DPMA historical profile quoting Willstätter's autobiography [19].

C) **HUMAN REALITY DETAILS (8–15 bullets)**

- Born into a Jewish family in Karlsruhe (1872) and studied chemistry at Munich under Adolf von Baeyer [20]. He remained at Munich 15 years before moving to Zurich (ETH) in 1905. — *Source:* DPMA/Nobel biography [20].

- Married Sophie Leser in 1903; she died in 1908. They had one son (Ludwig) and one daughter (Ida Margarete) [21]. (He never remarried.) — *Source:* Nobel biographical sketch [21].

- He regarded his 1905–1912 professorship at ETH Zurich as "the best and most significant" period of his career, despite feeling personally lonely there [22]. — *Source:* Nobel biographical sketch [22].

- Willstätter acknowledged his Jewish origins openly (unlike some friends) but considered faith a private matter [9]. His resignation letter in 1924 was framed in polite, formal tone (as above) rather than defiance. — *Source:* DPMA history [9].

- He was a highly sought-after consultant: during his career he advised major chemical firms (Bayer, Boehringer, Merck, Sandoz) and held several lucrative patents [23]. His industrial income made him financially secure enough to retire early. — *Source:* Trauner (2015) [23].

- Wealth permitted a cultured lifestyle: in the 1920s he built a modern villa in a fashionable Munich district (near Thomas Mann's home) to house his vast personal library and art collection [24]. — *Source:* Trauner (2015) [24].

- During WWI he contributed to the war effort (refusing poison gas work) by co-inventing a protective gas mask filter (1916); he was awarded the Iron Cross Second Class for it [25]. — *Source:* DPMA biography [25].

- After his 1924 retirement, he continued private research at home. He ran a small lab operation largely by telephone with his assistant Margarete Rohdewald (who worked from Munich even after he moved) [26]. — *Source:* Trauner (2015) [26].

- Colleagues noted his integrity: for example, psychiatrist Emil Kraepelin (a fellow professor) publicly defended him in 1924, but Willstätter refused to remain under discrimination [10]. — *Source:* DPMA account [10].

- In late 1938 he wrote “The downfall of humanity in Germany did not take place until after the end of my professional activity” (reflecting on Nazi atrocities) . — *Source:* DPMA biography quoting his reflections .[27]

- The night after Kristallnacht (Nov 1938), Gestapo came for him, but he was working in his garden and avoided capture .[28] He later recalled shivering for nights at his desk, ready for them to return, until fleeing. — *Source:* Trauner (2015) .[15]

- With student assistance (Arthur Stoll and others) he fled Germany in early 1939. He later remarked on emigration: *“I left my beloved German homeland, which had been everything to me…”* . — *Source:* DPMA historical profile .[19]

- He spent his final years (1939–1942) in a modest villa “Villa Eremitaggio” in Muralto, Switzerland, writing his memoir *Aus meinem Leben*. The book was published posthumously in 1949 (English 1965) . — *Source:* Nobel biographical sketch .[29][29]

D) **SCIENCE CONNECTION (Intro Biochem)**

- **Best-fit topic:** Photosynthetic pigments (plant metabolism) and enzymes.
- **Precise contribution:** Willstätter chemically characterized chlorophyll and flower pigments, showing chlorophyll contains a magnesium-porphyrin ring and isolating anthocyanins; he also pioneered purification techniques in enzymology (though controversially, he doubted enzymes were proteins).
- **Bridge explanation:** Chlorophyll is the key light-absorbing molecule in photosynthesis. Willstätter showed that chlorophyll’s structure is very similar to hemoglobin’s porphyrin ring, except magnesium (Mg^{2+}) replaces iron .[6][5] This explains why chlorophyll absorbs visible light: the porphyrin ring is a strong light catcher. In intro biochemistry, students learn that chlorophyll and hemoglobin are both tetrapyrroles (porphyrins) with a central metal ion, underlining a common theme in bioinorganic chemistry (porphyrins as cofactors).
  Specifically, chlorophyll can be viewed as a cofactor (like a coenzyme) that carries out the energy capture in photosynthesis. Willstätter’s work makes the point that swapping a metal in a ring changes function: Mg in chlorophyll lets plants trap sunlight, whereas Fe in hemoglobin lets animals carry oxygen.
- **Scientist’s analogy:** Willstätter himself noted this parallel with blood pigment: *“the chlorophyllins are magnesium compounds… as haemoglobin contains iron”* .[6] He also explained the red/blue color shift in anthocyanins by pH, akin to a pH indicator changing color. (His view was later refined, but the idea illustrates how biochemical molecules can change form with pH.)
- **Sources:** Nobel Prize facts ; Nobel lecture ; MPIC KWI history .[8][6][5]

E) **COMPLICATIONS OR CONTROVERSIES**

- **Enzymes and Protein Debate:** *What's agreed:* Willstätter's lab purified peroxidase and chlorophyllase to high purity and found no detectable protein content [30], leading him to doubt that enzymes were proteins. *What's disputed:* The nature of enzymes – by 1926 James Sumner crystallized urease and showed it is a protein, demonstrating that enzymes *are* proteins [30]. Thus the initial claim (enzyme=not protein) was overturned. *Sources:* Willstätter & Weber (1926) studies cited in Trauner [30] versus Sumner 1926 (cited in same source). Both perspectives appear in biochemical reviews [30] (Willstätter's view vs. Sumner's discovery).

- **Tropinone Synthesis Credit:** *What's agreed:* Willstätter first synthesized tropinone (1901) and cocaine [1]. *What's disputed:* Robinson's 1917 "double Mannich" one-pot synthesis became more famous [31]. Willstätter claimed he had conceived a similar route to tropinone and even patented a 5 cocaine synthesis, suggesting Robinson "stole" the idea [32]. The dispute is one of priority/recognition in organic synthesis. *Sources:* Mukesh Doble (2007) on Willstätter's 1901 work [1]; Trauner (2015) on Robinson vs. Willstätter [32].

- **Anthocyanin Color Explanation:** *What's agreed:* Willstätter isolated cornflower cyanin and its rose [3] counterpart. *What's disputed:* The cause of their different colors. Willstätter attributed it solely to pH differences (acidity). Peers (including his collaborator Everest) later argued that metal complexes or co-pigments were needed. (Modern chemistry finds that a ferric/magnesium complex gives the cornflower its true blue.) Thus, the simplistic pH-only theory was contested. *Sources:* Takeda (2006) notes Willstätter's pH conclusion [3]; Everest's rebuttal is noted in the same review (not directly cited here).

F) **"NOT ON WIKIPEDIA" FINDS (5+)**

- **Widespread Consulting and Wealth:** Willstätter was a paid consultant to industry (Bayer, Boehringer, Merck, Sandoz) and held lucrative patents, which made him financially independent [23]. *Why distinctive:* His wealth and industry ties (rare for academics then) explain how he could retire early and live as a "gentleman of leisure." *Source:* Trauner 2015 [23].

- **Turned Down Prestigious Jobs:** He declined several high-profile positions: to succeed Emil Fischer in Berlin, to head Siemens Research labs, to lead a Rockefeller Institute in Madrid (at Cajal's invitation), and to join the University of Chicago [24]. *Why distinctive:* Shows his fame and choices – he freely refused world-class offers to remain in Munich and work on his own terms. *Source:* Trauner 2015 [24].

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Munich Villa and Library: In Munich Willstätter built a modern villa in a fashionable neighborhood (Thomas Mann was a neighbor) to house his vast library and art collection [26]. *Why distinctive:* Illustrates his cultured personal life; the villa later became a Nazi target. *Source:* Trauner 2015 [26].

- **Work by Telephone (1920s):** After retiring, he did research from home, collaborating by phone. For example, he ran a small project with his assistant Margarete Rohdewald through telephone reports [26]. *Why distinctive:* Unusual remote-lab work in the 1920s – an early example of distance collaboration. *Source:* Trauner 2015 [26].

- **Autobiography Praised:** His memoir *Aus meinem Leben* (1949) has been called "one of the most beautifully written scientific autobiographies" [33]. *Why distinctive:* Highlights that Willstätter was not only a chemist but also a notable writer; his first-hand account is highly readable and valued by historians. *Source:* Trauner 2015 [33].

- **Last-minute Escape:** After Kristallnacht, he avoided arrest by sheer luck (was in his garden when Gestapo came) [17]. He later wrote chillingly of waiting in vain for them to return: *"the following three early mornings I sat at my desk… but they didn't come… I had to emigrate."* [16]. *Why distinctive:* Provides vivid personal color to his final flight; such first-person detail is absent from Wikipedia. *Sources:* Trauner 2015 [17]; DPMA profile [16].

- **"Toleranza per riposo" Motto:** He described finding *"toleranza per riposo"* (Italian for "tolerance for rest") in his exile home at Villa Eremitaggio in Muralto [19]. *Why distinctive:* This poetic phrase (from the inscription on the Swiss villa) reflects his longing for peace; it's a personal touch not on Wikipedia. *Source:* DPMA profile [19].

G) **SOURCES CONSULTED (Full citations)**

- R. Willstätter – Nobel Lecture: *On Plant Pigments*, June 3, 1920, Nobel Prize in Chemistry 1915 (Nobelprize.org archives) [6].
- "The Nobel Prize in Chemistry 1915 – Facts" and "Biographical" (NobelPrize.org) [4][8].
- Dirk Trauner, "Richard Willstätter and the 1915 Nobel Prize in Chemistry," *Angew. Chem. Int. Ed.* 54 (2015): 11910–11916 [32][24].
- Max-Planck-Institut für Chemie, "Chronicle of the Directors" (MPI Berlin history) [5][12].
- Mukesh Doble & Anil Kumar Kruthiventi, *Green Chemistry and Engineering* (Wiley, 2007), ch. on tropinone [1].
- Kosaku Takeda, "Blue Metal Complex Pigments in Flowers," *Proc. Japan Acad.* 82 (2006): 142–154 [3].
- Deutsches Patent- und Markenamt (DPMA) Milestones: "150th birthday of Richard Willstätter" (historical profile, 2025) [9][20].

- *From My Life* by Richard Willstätter (memoir), ed. Arthur Stoll (Verlag Chemie, 1949) – referenced in Trauner and DPMA texts.
- [2] Oxford English references: Tropinone and cocaine synthesis (ChemicalBook encyclopedia) . • Various secondary sources on enzymology and history (e.g. Sumner enzyme crystallization reviews) cited in Trauner's bibliography.

H) **FLAGGED "MYTH-RISK" DETAILS**

No widely repeated anecdotes without documentation were identified. All dramatic claims (resignation letter, Gestapo escape, Nobel work) are backed by primary or scholarly sources. For instance, the story of [17] Willstätter hiding in his garden during Kristallnacht is well-sourced , not just hearsay. Therefore, we do not list any unverified "myth" claims.

LENS RECOMMENDATION

Lens Score Justification	
(Discovery) Eureka	3 Multiple well-documented discovery scenes (tropinone, anthocyanin, chlorophyll) with concrete experimental detail [1][3][6] .
(Setback) Failure	3 Strongly documented turning point (1924 resignation) with vivid details (letter text) and lasting career impact [9] .
Mentor	0 No single pivotal mentor/mentee story with scene-level detail was found.
Controversy	2 Some disputes (enzymes vs. protein, tropinone credit) exist but are not central in surviving narratives.
Human Detail	3 Rich personal details (resignation letter, exile quotes, Villa) that deeply color his story, all well-sourced.
Last Experiment	2 Late-life escape and memoir writing provide a "final act," but with less of a single climactic scene.

Primary recommended lens: *Failure (Setback)*. Willstätter's principled resignation in 1924 and later narrow escape from Nazi arrest create the most vivid, scene-worthy narratives with strong documentation. Scene #4 (Munich resignation) and #5 (Gestapo escape) exemplify this lens, showing how external forces changed [9] his life. Quote #3 ("I humbly ask… My decision is final") powerfully illustrates this turning point. *Cautions:* The letter quote is well-sourced; no significant myths appear.

Backup lens: *Eureka (Discovery)*. His story equally centers on major discoveries. Scenes #1–#3 (tropinone synthesis, pigment isolations) have clear causal narratives. An early quote for an opening could be the Nobel Lecture: "The intention behind my work was to establish the constitutional characteristics of…the [18] chlorophyll in particular" , which frames his drive. This lens ties to the discovery of chlorophyll's structure (Scene #3) as the key science story. (Controversy points are minor and well-documented.)

[1] [31] Tropinone - an overview | ScienceDirect Topics
https://www.sciencedirect.com/topics/chemistry/tropinone

[2] Tropinone - Wikipedia
https://en.wikipedia.org/wiki/Tropinone

[3] Blue metal complex pigments involved in blue flower color - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC4323046/

[4] [21] [22] [29] Richard Willstätter – Biographical - NobelPrize.org
https://www.nobelprize.org/prizes/chemistry/1915/willstatter/biographical/

[5] [12] Chronicle of the Directors
https://www.mpic.de/3537797/chronicle-since1912

[6] [7] [18] Richard Willstätter - Nobel Lecture
https://www.nobelprize.org/uploads/2018/06/willstatter-lecture.pdf

[8] Richard Willstätter – Facts - NobelPrize.org
https://www.nobelprize.org/prizes/chemistry/1915/willstatter/facts/

[9] [10] [11] [13] [16] [19] [20] [25] [27] DPMA | Richard Willstätter
https://www.dpma.de/english/our_office/publications/milestones/greatinventors/richardwillstaetter/index.html

[14] [15] [17] [23] [24] [26] [28] [32] [33] Richard Willsttter and the 1915 Nobel Prize in Chemistry
https://www2.chem.wisc.edu/deptfiles/chem345-gellman/Sp16/Willstatter%20Angew%2015.pdf

[30] Richard Willstätter and the 1915 Nobel Prize in Chemistry
https://www.researchgate.net/publication/281145094_Richard_Willstatter_and_the_1915_Nobel_Prize_in_Chemistry

Profile

Richard Willstätter (1872–1942) was a German chemist whose work played a foundational role in shaping modern biochemistry by revealing how biological pigments and enzymes function at the molecular level. Working at the intersection of organic chemistry and biology, Willstätter

helped move the study of life away from vague “vital forces” toward precise chemical structures, a shift that was essential for the development of biochemistry as a scientific discipline.

Willstätter is best known for his groundbreaking research on chlorophyll, the green pigment responsible for photosynthesis. Before his work, chlorophyll was treated as a poorly understood plant extract. Between 1906 and 1914, while working at the Kaiser Wilhelm Institute in Berlin, Willstätter and his collaborators showed that chlorophyll is not a single substance but a mixture of two closely related compounds, now known as chlorophyll *a* and chlorophyll *b*. Through careful chemical degradation and purification, he demonstrated that chlorophyll contains a porphyrin ring with a magnesium ion at its center. This discovery was especially important because it revealed a striking parallel between plant and animal biology: hemoglobin, the oxygen-carrying molecule in blood, also contains a porphyrin ring, but with iron instead of magnesium.

This structural insight explained how chlorophyll absorbs visible light and converts it into chemical energy during photosynthesis. By showing that chlorophyll’s function could be understood through its molecular structure, Willstätter helped establish a core principle of biochemistry—that biological activity arises from chemical form. For this work, he was awarded the 1915 Nobel Prize in Chemistry “for his researches on plant pigments, especially chlorophyll.”

Willstätter’s influence extended beyond photosynthesis into enzymology, another central topic in introductory biochemistry. At a time when the chemical nature of enzymes was still debated, he insisted that enzymes must be purified to homogeneity before conclusions could be drawn about their composition. His laboratory developed rigorous purification techniques for enzymes such as

peroxidase and chlorophyllase. Although Willstätter concluded that enzymes were not proteins—a view later disproven by James Sumner's crystallization of urease—his emphasis on purity and experimental rigor shaped how enzyme research was conducted. Even when his conclusions were revised, his methods became standard practice.

Willstätter's life also demonstrates how scientific careers can be shaped by social and political forces. In 1924, amid rising antisemitism at the University of Munich, posters appeared declaring that "no German youth" should be taught by a Jewish professor. In response, Willstätter resigned his professorship in protest, formally requesting dismissal and stating that his decision was final. Despite international recognition, he withdrew from academic life in Germany.

Following Kristallnacht in 1938, Willstätter narrowly escaped arrest by the Nazi regime and fled to Switzerland, leaving behind his home, library, and Nobel Prize medal. He spent his final years in quiet exile, reflecting on his life and writing an autobiography.

For students of biochemistry, Willstätter's legacy lies not only in his discoveries but in his approach to science. His work shows that understanding life requires understanding molecules—and that careful chemical analysis can reveal the hidden mechanisms behind biological processes.